United States Patent	12409938
Kind Code	B2
Date of Patent	September 09, 2025
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Seat arrangements, in particular for an aircraft cabin

Abstract

A seat arrangement for an aircraft cabin includes a first column of seats extending on one side of an aisle, a second column of seats extending on an opposite side of the aisle, and a door. The seats of the second column are staggered with respect to the seats of the first column. The door can move between a stowed position and a deployed position in which the door closes an upper space located above a console of the first column. The upper space is defined by a shell associated with the console and by a shell associated with a front seat. A lower space can extend between the front end of the console and the shell of the front seat being left free. The lower space communicates with the upper space to form an access passage when the door is in the stowed position.

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Appl. No.:	**17/627486**
Filed (or PCT Filed):	**July 08, 2020**
PCT No.:	**PCT/EP2020/069283**
PCT Pub. No.:	**WO2021/008974**
PCT Pub. Date:	January 21, 2021

Prior Publication Data

Document Identifier	**Publication Date**
US 20220242572 A1	Aug. 04, 2022

Foreign Application Priority Data

FR	1908144	Jul. 18, 2019

Publication Classification

Int. Cl.: **B64D11/06** (20060101)

U.S. Cl.:

CPC **B64D11/0606** (20141201);

Field of Classification Search

CPC: B64D (11/0601); B64D (11/0602); B64D (11/0604); B64D (11/0606); B64D (11/0023); B64D (11/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10237511	12/2018	Garing	N/A	H04N 5/7408
11319072	12/2021	Henshaw	N/A	B64D 11/0601
2013/0038103	12/2012	Scott	297/248	B60N 2/3025
2013/0160240	12/2012	Kenny	N/A	N/A
2013/0241247	12/2012	Wallace	297/118	B64D 11/0604
2016/0272323	12/2015	Carlioz	N/A	B64D 11/0602
2017/0172301	12/2016	Janzen	N/A	A47B 88/493
2018/0281963	12/2017	Dowty	N/A	B64D 11/0606

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3033918	12/2017	CA	B64D 11/0601
3225548	12/2016	EP	N/A
3225551	12/2016	EP	N/A
3059951	12/2017	FR	N/A
2548901	12/2016	GB	N/A
2012140515	12/2011	WO	N/A
2018033599	12/2017	WO	N/A

WO-2018093825	12/2017	WO	B60N 2/005
WO-2018184778	12/2017	WO	B64D 11/0023
WO-2019179638	12/2018	WO	B64D 11/0015

OTHER PUBLICATIONS

International Patent Application No. PCT/EP2020/069283, International Search Report (with English translation) and Written Opinion, dated Sep. 16, 2020. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is a national stage application of International Patent Application PCT/EP2020/069283, filed on Jul. 8, 2020 and titled “SEAT ARRANGEMENTS, IN PARTICULAR FOR AN AIRCRAFT CABIN,” which is related to and claims priority to France Patent Application No. 1908144, filed on Jul. 18, 2019, both of which are hereby incorporated by reference in their entireties.
FIELD OF INVENTION
(2) The present invention relates to an arrangement of seats, especially for an aircraft cabin.
BACKGROUND
(3) The seats of the ‘business class’ and ‘first class’ type offer passengers different positions of comfort, from a ‘seating’ position to a ‘lying’ position, in which the seat defines a substantially horizontal lying surface so that the passenger can lie down.
(4) Intermediate positions of comfort are also available, such as the ‘relaxing’ position in which a back of the seat is strongly inclined. Generally, these intermediate positions are obtained by inclining the pivoting backrest about a horizontal axis and perpendicular to an axis of extension of the seat. The passenger can then remain in the seat during transitions among the different positions.
(5) Generally, the seat comprises in particular the backrest and a seating surface and is likely to include a legrest and/or a footrest, which can be fixed or linked to a kinematics of the seat.
(6) Arrangements of seats for aircraft cabin of ‘business class’ and ‘first class’ type include a passage, disposed between two seat units arranged one behind the other along a longitudinal axis of the aircraft cabin, allowing direct access to an aisle for all passengers. Passengers can thus easily access to the aisle, especially when the seat is in an extended position.
(7) Therefore, when a passenger or a crew member is walking in the aisle located laterally to the seat, s/he is likely to be in direct visual contact with the passenger in the seat. Such a situation can bring a feeling of discomfort to the passenger in the seat, in that s/he feels a lack of privacy.
(8) Document US2018/0281963 describes a device for closing a passage between two seat units arranged one behind the other. This closing device comprises a door mounted so as to slide between a stored position, in which the door leaves the passage unobstructed, and an extended position, in which the door closes the passage. Such a configuration makes it possible to define an enclosed interior space around the seat, providing thereby a feeling of privacy to the passenger.

According to such a configuration of the state of the art, the passenger in the seat is not in direct sight of other passengers and/or crew members walking in the aisle. Documents EP3225548 and GB2548901 describe a similar closing device.
(9) However, it is not easy to integrate such doors into a configuration of seats including at least two columns in which the seats are staggered. In fact, certain passages for getting access to a seat in the furthest column from the aisle do not allow the integration of a conventional door extending almost along the entire height of the seat module, because it is difficult to comply with the safety constraints imposed by aeronautical standards due to the short length of the passage.
SUMMARY
(10) The invention aims at effectively remedying this drawback by providing an arrangement of seats, in particular for an aircraft cabin, comprising: at least a first column of seats extending from one side of an aisle and comprising seats arranged one behind the other in alternation with consoles, a console being provided with a foot zone for receiving the feet of a passenger in a corresponding rear seat in the first column, at least a second column of seats extending from a side opposite to the aisle and comprising seats arranged one behind the other in alternation with consoles, a console being provided with a foot zone for receiving the feet of a passenger in a corresponding rear seat in the second column, the seats in the second column being staggered with respect to the seats in the first column, said arrangement comprising at least one door movable between a stored position and a deployed position in which the door closes an upper space above a console in the first column, said upper space being delimited on the one hand by a shell associated with said console and on the other hand by a shell associated with a front seat, a lower space extending between a front end of the console and the shell of the front seat being left free, the lower space being in communication with the upper space to form an access passage between the aisle and a seat in the second column when the door is in the stored position.
(11) The invention thus makes it possible to guarantee the privacy of the passengers in a configuration of seats in a staggered arrangement along two columns, while respecting the safety regulations due to the lower space left free between the console and the front seat, allowing thereby evacuation in case of emergency.
(12) According to one embodiment, the interior space left free has a length of at least 9 inches (22.86 cm).
(13) According to one embodiment, the lower space left free has a height between 25 inches (63.50 cm) and 30 inches (76.20 cm).
(14) According to one embodiment, the door and the upper space have a length between 15 inches (38.10 cm) and 20 inches (50.80 cm).
(15) According to one embodiment, the door and the upper space have a height between 14 inches (35.56 cm) and 20 inches (50.80 cm).
(16) According to one embodiment, said arrangement comprises at least a second door movable between a stored position and an extended position in which the door closes a space extending between a shell of a seat and a rear end of a console in the first column, this space forming an access passage between the aisle and a seat in the first column, when the door is in the stored position.
(17) According to one embodiment, at least one space extending between a console and a seat in the first column may be without a door.
(18) According to one embodiment, in particular at one end of a column of seats, a premium space comprises a bed space and/or an increased seating space and/or a movie space.
(19) According to one embodiment, the door is arranged between a portion of a shell and a fixed support carrying a guiding slide, a rack, components of an automatic deployment system, as well as components of a system for locking and unlocking the door.
(20) According to one embodiment, the door comprises a carriage mounted so as to be mobile in translation along a guiding slide in order to allow the door to move from one position to another.

(21) According to one embodiment, the arrangement comprises a system for locking and unlocking the door relative to the carriage.
(22) According to one embodiment, the door comprises a second carriage provided with a damping pinion for damping a movement of the door and cooperating with a rack.
(23) According to one embodiment, the second carriage comprises a locking and unlocking system capable of assuming a locked state in which said system maintains the second carriage in a high position so that the teeth of the damping pinion cooperate with the rack, and an unlocked state allowing a movement of said second carriage to a low position so that the teeth of the damping pinion are released from the rack.
(24) According to one embodiment, said arrangement comprises a control device, in particular including cables, capable of controlling a change of state of at least one locking and unlocking system.
(25) According to one embodiment, said arrangement comprises a system for locking the door in the stored position which can be actuated by a crew member.
(26) According to one embodiment, said arrangement comprises a system for automatically deploying the door from a stored position to a deployed position.
(27) The object of the invention is also an aircraft comprising an arrangement of seats as defined above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The invention will be better understood and other characteristics and advantages will appear by reading the following detailed description, which includes embodiments given for illustrative purposes with reference to the accompanying figures, presented as way of non-limiting examples, which may serve to complete the understanding of the present invention and the description of its implementation and eventually contribute to its definition, wherein:
(2) FIG. **1** is a top view of an arrangement of seats according to the present invention;
(3) FIG. **2** is a top view of an aircraft cabin incorporating an arrangement of seats according to the present invention;
(4) FIG. **3** is a side view illustrating the two types of doors which can be used in the arrangement of seats according to the invention;
(5) FIG. **4** is a side view showing a small door for getting access to a seat in the furthest column from the aisle in the stored position;
(6) FIG. **5** is a side view showing a large door for getting access to a seat in the closest column to the aisle in the stored position;
(7) FIG. **6** is a perspective view of a seat in the furthest column from the aisle in the extended position;
(8) FIG. **7** is a perspective view illustrating the integration of a door and its operating mechanism between a shell and its support;
(9) FIG. **8** is a detailed perspective view of a guiding carriage for the door according to the invention;
(10) FIG. **9** is a perspective view of the system for locking and unlocking the guiding carriage for the door according to the invention;
(11) FIG. **10** shows a perspective view of the upper carriage of the door according to the invention provided with a damping pinion cooperating with a rack;
(12) FIGS. **11***a* and **11***b* show a perspective view of a locking system for the door according to the invention respectively in the released position and in the locked position;
(13) FIG. **12** shows a side view of a system for automatically deploying a closing door for a space

according to the invention.

DETAILED DESCRIPTION

(14) It should be noted that, in the figures, the structural and/or functional elements common to the different embodiments may have the same references. Thus, unless otherwise stated, such elements have identical structural, dimensional and material properties.

(15) FIG. **1** shows an arrangement of seats **10**, in particular for an aircraft cabin, comprising at least a first column C**1** of seats **11**.**1** extending from the side of an aisle **12**. The first column C**1** comprises seats **11**.**1** arranged one behind the other in alternation with consoles **13**.**1**. A console **13**.**1** is provided with a foot zone **15** for receiving the feet of a passenger **14** in a corresponding rear seat **11**.**1** in the first column C**1**.

(16) At least a second column C**2** of seats **11**.**2** extends from an opposite side of the aisle **12**. The second column C**2** comprises seats **11**.**2** arranged one behind the other in alternation with consoles **13**.**2**. A console **13**.**2** is provided with a foot zone **15** for receiving the feet of a passenger **14** in a corresponding rear seat **11**.**2** in the second column C**2**.

(17) The seats **11**.**1** in the first column C**1** and the seats **11**.**2** in the second column C**2** are turned in the same direction D which may correspond to the direction of movement of the aircraft. As a variant, the orientation direction of the seats **11**.**1**, **11**.**2** may correspond to an opposite direction to the movement of the aircraft, or any other direction.

(18) A column of seats C**1**, C**2** extends longitudinally inside the aircraft. A column of seats C**1**, C**2** can extend along a longitudinal axis of the aircraft cabin X**1** or along an axis parallel to the longitudinal axis of the aircraft cabin X**1**, as shown in FIG. **2**.

(19) It should be noted that, in the remainder of the description, the terms ‘front’ and ‘rear’ used in relation to the term ‘seat’ or ‘console’ locally define a relative position of the seat or of the console inside a column C**1**, C**2** and do not refer in any way to an orientation of the seat or the console inside the aircraft cabin. In other words, the expression ‘front seat’ means that the front seat is positioned directly in front of another seat inside the column C**1**, C**2**, while the expression ‘rear seat’ means that the seat is positioned directly behind another seat inside the column C**1**, C**2**.

(20) Inside a column, each seat **11**.**1**, **11**.**2** has an axis X**2** which may be parallel to the longitudinal axis of the aircraft cabin X**1**. Alternatively, the axis X**2** of a seat **11**.**1**, **11**.**2** may form an angle, for example between 0 and 20 degrees, with respect to the longitudinal axis of the aircraft cabin X**1**. The axis X**2** of a seat **11**.**1**, **11**.**2** corresponds to an intersection of a vertical plane of symmetry of the seat **11**.**1**, **11**.**2** with a corresponding horizontal plane.

(21) The column C**1** in the arrangement of seats **10** may be arranged on the side of the aisle **12**, while the second column C**2** may be arranged on the side of the fuselage **16**. Thus, in the arrangement in FIG. **2**, one distinguishes a first set E**1** of two columns of seats C**1**, C**2** and a second set E**2** of two columns of seats C**1**, C**2**, symmetrical to the longitudinal axis of the aircraft cabin X**1**. Each set E**1**, E**2** has a column of seats C**2** arranged on the side of the fuselage **16**. There are also two central sets E**3**, E**4** in two columns C**1**, C**2** symmetrical to the longitudinal axis of the aircraft cabin X**1**.

(22) A seat **11**.**1**,**11**.**2** is convertible between a ‘seating’ position, corresponding to the position used in particular during the stopping, take-off, and landing phases of the aircraft, and a ‘lying’ position’, in which the seat **11**.**1**, **11**.**2** defines a substantially horizontal plane for the passenger **14** in order to allow her/him to rest, as shown in FIG. **6**. In particular, the ‘seating’ position and the ‘lying’ position are two extreme positions of the seat **11**.**1**, **11**.**2**. According to alternative embodiments, the seat **11**.**1**, **11**.**2** may also assume intermediate positions, called relaxing positions, between these two extreme positions.

(23) Preferably, the seats **11**.**2** in the second column C**2** and the seats **11**.**1** in the first column C**1** are in a staggered arrangement, that is to say the seats **11**.**1**, **11**.**2** are not laterally aligned from one column relative to the other.

(24) In each column of seats C**1**, C**2**, a console **13**.**1**, **13**.**2** is provided with a foot zone **15** for

receiving the feet of a passenger **14** in a corresponding rear seat **11.1**, **11.2**, in particular when the seat **11.1**, **11.2** is in a lying position. Thus, the opening of the foot zone **15** is directed towards the seat **11.1**, **11.2** immediately behind the corresponding console **13.1**, **13.2**.
(25) A given console **13.1**, **13.2** is arranged near a seat **11.1**, **11.2** of the other column so as to define a space for storing and/or placing objects. A console **13.1**, **13.2** includes for this purpose a flat upper wall **18** on which the passenger **14** in the adjacent seat **11.1**, **11.2** can in particular place objects. The upper wall **18** can thus serve as a table, a working surface, a dining table, a cocktail table or any other usable surface for a passenger **14** in the adjacent seat **11.1**, **11.2** in the other column. The console **13.1**, **13.2** may also include storage spaces such as magazine rack, drawers, or the like.
(26) A console **13.1**, **13.2** and an adjacent seat **11.1**, **11.2** in the other column may form a seat module MD, MG. There is a right seat module MD in which the console **13.2** is arranged on the right of the corresponding seat **11.1** and a left seat module MG in which the console **13.1** is arranged on the left of the corresponding seat **11.2**. It is thus possible to form the set of two columns C**1**, C**2** by alternately arranging a right module MD and a left module MG one behind the other.
(27) A seat **11.1**, **11.2** is associated with a privacy shell **20** for the passenger **14**. A console **13.1**, **13.2** is also associated with a privacy shell **21**. The shells **20** and **21** of a seat **11.1**, **11.2** and a console **13.1**, **13.2** of the same module MD, MG can be made in one piece.
(28) As can be seen in FIGS. **3**, **4**, **5**, and **6**, at least one door **23.1** is movable between a stored position and a deployed position in which the door **23.1** closes an upper space **24** above a console **13.1** in the first column C**1**. This upper space **24** is delimited on the one hand by the shell **21** associated with the console **13** and on the other hand by the shell **20** associated with the front seat **11.1**.
(29) A lower space **25** extending between the front end of the console **13.1** and a shell **20** of the front seat **11.1** is left free, that is to say this lower space **25** has no door. The lower space **25** is preferably left free along the entire height of the console **13.1**. The lower space **25** is narrower than the upper space **24**. The lower space **25** is in communication with the upper space **24** so as to form an access passage between the aisle **12** and a seat **11.2** in the second column C**2** when the door **23.1** is in the stored position.
(30) Thus, the door **23.1** is arranged in the free space above the console **13.1**, in particular above a surface of the console **13.1** forming a table. This makes it possible to facilitate the integration of the door **23.1**, which would otherwise reduce, due to its thickness, the available width of the seat **11.2** and/or the console **13.1**. In addition, this makes it possible to maximize the space under the first door **23.1** associated with a seat **11.2** remote from the aisle **12** for an evacuation in the event of an emergency. As a passenger can thus easily pass under a door **12.1**, it is not necessary to carry out an evacuation test for the doors **12.1**.
(31) According to an exemplary embodiment, the lower space **25** left free has a length L**1** of at least 9 inches (22.86 cm), and a height L**2** between 25 inches (63.50 cm) and 30 inches (76.20 cm). The door **23.1** and the upper space **24** have a length L**3** between 15 inches (38.10 cm) and 20 inches (50.80 cm) and a height L**4** between 14 inches (35.56 cm) and 20 inches (50.80 cm).
(32) In addition, at least a second door **23.2** is movable between a stored position and an extended position in which the door **23.2** closes a space **27** extending between a shell **20** of a seat **11.1** and a rear end of a console **13.1** in the first column C**1**. This space **27** forms an access passage between the aisle **12** and a seat **11.1** in the first column C**1**, when the door **23.2** is in the stored position.
(33) The first door **23.1** is smaller than the second door **23.2**. In particular, the first door **23.1** has a height L**4**, measured in a vertical direction, inferior to the height of the second door **23.2**.
(34) When a door **23.1**, **23.2** is in the deployed position, the corresponding access passage between the aisle **12** and a seat **11.1** in the first column C**1** or a seat **11.2** in the second column C**2** is closed by the door **23.1**, **23.2**. Consequently, the passenger **14** in the seat **11** is separated from any

passenger and/or crew member walking in the aisle **12** by the door **23.1**, **23.2** and the privacy shells **20** and **21**. The passenger **14** is then seated in a privacy zone defined by the enclosed space thus created.
(35) Some spaces between consoles **13.1** and seats **11.1** in the first column C**1** may however be without a door **23.1**, **23.2**. This may be in particular the case for bottom-of-the-range seats **11.1**, **11.2** compared to those associated with a closing door **23.1**, **23.2**.
(36) Furthermore, a premium space EP may be provided, in particular at one end of a column of seats C**1**, C**2**. This premium space EP may include a bed space and/or an increased seating space and/or a movie space.
(37) As shown in FIG. **7**, a door **23.1** is disposed between a portion of a shell **21** and a fixed support **28** carrying a guiding slide **31**, a rack **40**, components of an automatic deployment system **52**, as well as components of a system **33** for locking and unlocking the door **23.1** described in more detail below.
(38) As can be seen in FIG. **8**, a door **23.1**, **23.2** comprises a carriage **30** mounted so as to be mobile in translation along a guiding slide **31** in order to allow the door **23.1**, **23.2** to move from the stored position to the deployed position and vice versa.
(39) A system **33** enables the door **23.1**, **23.2** to be locked and unlocked if necessary relative to the carriage **30**. According to an exemplary embodiment, the locking and unlocking system **33** comprises a rotary cam **34** mounted on the door **23.1**, **23.2**, as shown in FIG. **9**. The cam **34** is rotatably movable between a locking position in which the cam **34** abuts against one end of a release groove **35** and an unlocked position in which the axis of the cam **34** is aligned with the groove **35**, so as to be able to slide inside the groove **35**. This makes it possible to ensure a relative movement of the door **23.1**, **23.2** with respect to the carriage **30**.
(40) A sliding system **36**, for example with a rib and a corresponding groove, may make it possible to facilitate the movement of the door **23.1**, **23.2** relative to the carriage **30** to ensure the clearance of the access passage.
(41) As illustrated by FIG. **10**, the door **23.1**, **23.2** comprises a second carriage **38** provided with a damping pinion for damping a movement of the door **23.1**, **23.2** and cooperating with a rack **40** mounted on the fixed support **28**. The damping pinion **39** may have the form of a friction pinion of a known type.
(42) The second carriage **38** comprises a second system **42** for locking and unlocking the second carriage **38** and capable of assuming a locked state in which said system **42** maintains the second carriage **38** in a high position so that the teeth of the damping pinion **39** cooperate with the rack **40**, and an unlocked state allowing movement of said second carriage **38** to a low position so that the teeth of the damping pinion **39** are released from the rack **40**. The system **42** may include for example a cam **43**, the displacement of which relative to a notch **44** makes it possible to pass from one position to another.
(43) A control device **45**, in particular including cables, is able to control a change of state of at least one locking and unlocking system **33**, **42** via a displacement of at least one cam **34**, **43** of the corresponding system. The movement of the cam **34**, **43** may be controlled in particular by actuating an unlocking member, such as a button or a lever, integrated on one side of the door **23.1**, **23.2**.
(44) Furthermore, as shown in FIGS. **11***a* and **11***b*, a locking system **47** allows the door **23.1**, **23.2** to be locked in the stored position. This locking system **47** is intended for use by a crew member in order to ensure in particular that the door **23.1**, **23.2** is in the stored position when the seat **11.1**, **11.2** is in the TTL position (Taxi, Take-off, Landing) during the stopping, take-off and landing phases of the aircraft.
(45) The locking system **47** comprises a linear electric actuator **48** capable of moving a rod **49** in translation between a released position in which the rod **49** is released from a corresponding housing **50** provided in a portion of the first carriage **30** (cf. FIG. **11***a*) and a blocking position in

which the rod **49** penetrates inside the housing **50** of the carriage **30** to prevent the door **23.1**, **23.2** from moving (cf. FIG. **11***b*), under normal operating conditions.
(46) The electric actuators **48** can be controlled by a centralized system allowing the crew members to block all of the doors **23.1**, **23.2** in the stored position, in a single control operation.
(47) Advantageously, an automatic deployment system **52** for the door **23.1**, **23.2** is shown in FIG. **12**. The automatic deployment system **52** comprises an elastic strap **53** and a plurality of guiding means **54** for the strap **53**, such as pulleys, arranged between a first end of said strap **53** fixed on the door **23.1**, **23.2** and a second end of said strap **53** fixed on the fixed support **28**. An action by the passenger **14** on a control button **56** on the side of the seat **11.1**, **11.2** allows the strap **53** to be released and the door **23.1**, **23.2** to pass automatically in the deployed position.
(48) Of course, the different characteristics, variants and/or embodiments of the present invention can be associated with each other in various combinations insofar as they are not incompatible or mutually exclusive.
(49) Obviously, the invention is not limited to the embodiments described above and provided solely by way of example. It encompasses various modifications, alternative forms and other variants which a person skilled in the art may envisage in the context of the present invention and in particular any combination of the various operating modes described above may be taken separately or in combination.

Claims

1. An arrangement of seats for an aircraft cabin, the arrangement of seats, comprising: at least a first column of seats extending on one side of an aisle and comprising a first set of seats and consoles arranged one behind the other in a straight line, each console having a front end and a rear end, the rear end of each console being provided with a foot zone for receiving the feet of a passenger in a corresponding rear seat in the first column; at least a second column of seats located adjacent to the first column of seats and the side of a fuselage, the second column of seats extending parallel to the first column of seats, the second column of seats comprising a second set of seats and consoles arranged one behind the other in a straight line, each console having a front end and a rear end, the rear end of each console being provided with a foot zone for receiving the feet of a passenger in a corresponding rear seat in the second column, the seats in the second column being in a staggered arrangement with respect to the seats in the first column so that each seat of the first column is arranged adjacent to a console of the second column and each seat of the second column is adjacent to a console of the first column; at least one first door movable between a stored position and an extended position in which the at least one first door closes an upper space above a console in the first column of seats, said upper space being delimited on the one hand by a shell associated with the console and on the other hand by a shell associated with a front seat in the first column of seats; a lower space extending between the front end of each console in the first column of seats and the shell of the corresponding front seat being left free along the entire height of the corresponding console, the lower space having no door so that the lower space is left free when the first door is in the extended position; wherein the lower space in communication with the upper space forms an access passage between the aisle and a seat in the second column when the at least one first door is in the stored position; and at least one second door movable between a stored position and a deployed position in which the at least one second door closes a free space extending between a shell of the rear seat and the rear end of the console in the first column of seats, the free space forming an access passage between the aisle and the rear seat in the first column, when the at least one second door is in the stored position, wherein the at least one first door is smaller than the at least one second door, wherein the at least one first door or the at least one second door comprises a first carriage mounted so as to be mobile in translation along a guiding slide in order to allow the at least one first door or the at least one second door to move from one position to

another, wherein the at least one first door or the at least one second door comprises a second carriage provided with a damping pinion for damping a movement of the at least one first door or the at least one second door and cooperating with a rack, wherein the second carriage comprises a locking and unlocking system capable of assuming a locked state in which said system maintains the second carriage in a high position so that teeth of the damping pinion cooperate with the rack, and an unlocked state allowing said second carriage to move to a low position so that teeth of the damping pinion are released from the rack, wherein the locking and unlocking system includes a cam and a notch, a displacement of the cam relative to the notch allowing the second carriage to pass from one position to another.

2. The arrangement of seats according to claim 1, wherein the lower space left free has a length of at least 9 inches (22.86 cm).

3. The arrangement of seats according to claim 1, wherein the lower space left free has a height between 25 inches (63.50 cm) and 30 inches (76.20 cm).

4. The arrangement of seats according to claim 1, wherein the at least one first door and the upper space have a length between 15 inches (38.10 cm) and 20 inches (50.80 cm).

5. The arrangement of seats according to claim 1, wherein the at least one first door and the upper space have a height between 14 inches (35.56 cm) and 20 inches (50.80 cm).

6. The arrangement of seats according to claim 1, wherein at one end of the first column of seats and/or the second column of seats, a premium space comprises a bed space and/or an increased seating space and/or a movie space.

7. The arrangement of seats according to claim 1, wherein the at least one first door or the at least one second door is disposed between a portion of the shell of the console and a fixed support carrying a guiding slide, a rack, components of an automatic deployment system, as well as components of a system for locking and unlocking the at least one first door or the at least one second door.

8. The arrangement of seats according to claim 1, further comprising a system for locking and unlocking the at least one first door or the at least one second door relative to the carriage.

9. The arrangement of seats according to claim 8, further comprising a control device including cables, capable of controlling a change of state of at least one locking and unlocking system.

10. The arrangement of seats according to claim 1, wherein it comprises a locking system for the at least one first door or the at least one second door in the stored position, which can be actuated by a crew member.

11. The arrangement of seats according to claim 1, wherein it comprises an automatic deployment system for moving the at least one first door or the at least one second door from a stored position to a deployed position.

12. An aircraft comprising an arrangement of seats as defined in claim 1.

13. The arrangement of seats according to claim 1, wherein the at least one first door has a height measured in a vertical direction inferior to a height of the at least one second door.